Program 6: React Form with Validation and Toggle Password

6. Design and implement a React Form that collects user input for name, email, and password. Validate all fields before submission, ensure email format, enforce minimum password length, and add a 'Show Password' toggle.

Step-1: Create React Application

```
npx create-react-app react-form
```

Step-2: Update App.js

```
import { useState } from 'react';

export default function App() {
  const [f, setF] = useState({ n: '', e: '', p: '' });
  const [show, setShow] = useState(false);

  const valid = f.n && /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(f.e) && f.p.length >= 6;

  return (
    <div style={{ textAlign: 'center', marginTop: 40 }}>
      <h2>Form</h2>
      <input
        placeholder="Name"
        value={f.n}
        onChange={e => setF({ ...f, n: e.target.value })}
      /><br /><br />
      <input
        placeholder="Email"
        value={f.e}
        onChange={e => setF({ ...f, e: e.target.value })}
      /><br /><br />
      <input
        type={show ? 'text' : 'password'}
        placeholder="Password"
        value={f.p}
        onChange={e => setF({ ...f, p: e.target.value })}
      /><br /><br />
      <label>
        <input type="checkbox" onChange={() => setShow(!show)} />
        Show Password
      </label><br /><br />
      <button disabled={!valid} onClick={() => alert(JSON.stringify(f))}>
        Submit
      </button>
    </div>
  );
}
```